MAT 100 Precalculus

Homework 16

Basic Information

This assignment is due on Gradescope by **1:30 PM on Monday, November 9**.

Make sure you understand MHC honor code and have carefully read and understood the additional information on the class syllabus. I am happy to discuss any questions or concerns you have!

A component of this class is helping you understand *why* the math you use works the way it does. To that end, make sure you show all your work as you will be graded on the *process* you use, not just your final answer. And if a question asks you to explain *why* something is true, be sure to answer that part of the question in complete sentences.

Homework will be graded for correctness, and I will point out where mistakes or errors are made. Each homework will be graded *Excellent*, *Satisfactory*, or *Unsatisfactory*. The course rubric explains more about the distinctions of these categories.

Turn-In Problems

4.1: 8, 12
4.3: 2, 10, 18

Do NOT use Desmos or any other graphing program for the next 3. Solve the problems algebraically.

#6 Find all the asymptotes for $\frac{x^2}{x-1}$.

#7 Find the asymptotes for $\frac{1}{x}+\frac{1}{x-1}$. Suggestion: It may be helpful to combine into one fraction first.

#8 Find the asymptotes for $\frac{1}{x^2-6x+8}$.

#9 Use Demos to find the zeros of $x^3(4e^{4x})+3x^2e^{4x}$

Additional Problems (to do on your own, not to turn in)

4.1: 7, 11
4.3: 1, 9, 17